

Supplementary Material

Contagion: Per-Hop Prompt-Injection Survival in
LLM Agent Networks Does Not Compose

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This supplement contains the technical appendices referenced from the main paper (Appendices A–D). Cross-references of the form “Section n of the main paper” point to the main body; all other numbered items (equations, tables) are local to this supplement. Every number here is produced offline from stored results by `scripts/appendix_stats.py` and the analysis scripts listed in `REPRODUCE.md` (no model calls).

A Multiplicity Control for the Composition Test

Table 1 applies a Benjamini–Hochberg correction to the composition tests of Section 3.8 of the main paper, using the *default fresh-artefact protocol* for every cell at the common $n = 40$ grid. Only the Llama cell rejects H_0 ; it survives the correction at $q = 0.05$, while the four remaining non-degenerate cells—including DeepSeek, whose fixed-artefact deviation is a protocol artefact that disappears under the fresh protocol (Section 5.3 of the main paper)—stay non-significant. The Claude cell is degenerate ($\text{ASR} = \prod_i \hat{s}_i = 0$, $\text{sd}(\Delta) = 0$) and is excluded from the correction. At the higher $n = 200$ resolution the Nova cell also crosses into rejection (Section 5.10 of the main paper); we keep the correction at the common $n = 40$ grid so the five cells are compared at equal power.

Table 1: Benjamini–Hochberg over the $m = 5$ non-degenerate chain cells (fresh-artefact protocol). p -values are the bootstrap two-sided values from Section 3.8 of the main paper. Only the Llama rejection survives.

<i>rank</i>	<i>cell</i>	<i>p</i>	<i>BH thr. ($q=0.05$)</i>
1	Llama · none	0.0002	0.010
2	Nova · none	0.0840	0.020
3	qwen · none	0.3165	0.030
4	DeepSeek · none (fresh)	0.6300	0.040
5	qwen · paraphrase	0.7775	0.050

B Context-Clustered Interval on the Weak Edge

Section 5 of the main paper reports that a homogeneity test rejects exchangeability across benign task contexts on the weak Llama edge. Because that violates the i.i.d. assumption behind Wilson intervals, we recompute the edge rate with a cluster bootstrap that resamples *repeats* (seed replications), not individual trials. At a fixed temperature ($T = 0.7$, six repeats of twenty trials) the two intervals nearly coincide (Table 2): the design-effect widening is $0.99\times$, so within a temperature the Wilson interval is well calibrated. The overdispersion $\varphi = 2.93$ reported in Section 5 of the main paper is a *cross-temperature* artefact—temperature is a deliberate systematic factor, not trial noise—which is why we compute φ within a temperature.

C Threshold Sensitivity of the Floor Effect

A referee may ask whether the “resistant” verdict on the frontier model is an artefact of the compromise threshold $\tau_{\text{ASV}} = 0.9$. We re-score the single-hop obfuscation cells (Claude, $n = 30$

Table 2: Weak edge **a1** $\rightarrow$ **a2** (Llama, $T = 0.7$, $n = 120$): i.i.d. Wilson versus a repeat-clustered bootstrap (10^4 resamples).

<i>interval</i>	$\hat{s}$	95% <i>CI</i>	<i>width</i>
Wilson (i.i.d.)	0.242	[0.174, 0.326]	0.152
cluster bootstrap (by repeat)	0.242	[0.167, 0.317]	0.150

per cell) at $\tau \in \{0.7, 0.8, 0.9, 1.0\}$ directly from the stored ASV scores—a deterministic re-thresholding, no new model calls. The compromise rate is *invariant* across τ in every cell (Table 3): the no-defence baseline stays at 0.17 and redaction at 0.00 regardless of the threshold. The reason is that the ASV distribution is bimodal—of 180 trials, 158 score 0.0 and 14 score 1.0; the only eight intermediate trials all score ≤ 0.667 , i.e. below the lowest threshold tested. The floor effect is therefore a property of the model, not of where the judge’s cut is placed.

Table 3: Single-hop compromise rate versus τ_{ASV} (Claude, $n = 30$ per cell), re-scored offline from stored ASV. Rates are threshold-invariant.

<i>defence</i>	<i>form</i>	$\tau=0.7$	0.8	0.9	1.0
none	plain	0.17	0.17	0.17	0.17
none	spaced	0.17	0.17	0.17	0.17
none	split-word	0.13	0.13	0.13	0.13
redact	plain	0.00	0.00	0.00	0.00
redact	spaced	0.00	0.00	0.00	0.00
redact	split-word	0.00	0.00	0.00	0.00

D Recurrence Makes the Threshold Non-Vacuous

This appendix gives the full treatment of the recurrent (cyclic) case summarised in Section 5.8 of the main paper. We test a pattern real frameworks use: a manager broadcasting to workers that report back.

For a manager with w reporting workers, the infection matrix has $M[m \rightarrow w_j] = a_j$ and $M[w_j \rightarrow m] = c_j$, and its characteristic polynomial is $\lambda^{w-1}(\lambda^2 - \sum_j a_j c_j)$. Hence

$$\rho(M) = \sqrt{\sum_j a_j c_j} \quad (1)$$

for every w (checked numerically against the eigenvalues to machine precision on 320 random configurations). Two consequences follow. A feed-forward reading sets every $c_j = 0$, so $\rho = 0$ however strong the edges are—the vacuity of Section 3.8 of the main paper again, now with the feedback edges removed. And the system is supercritical exactly when $\sum_j a_j c_j > 1$; for symmetric edges $a_j = c_j = s$ this is $w s^2 > 1$. That is a *design rule* rather than a diagnostic: at $s = 0.5$ a manager needs five reporting workers to sustain an endemic injection, at $s = 0.7$ three, at $s = 0.9$ two.

Applied to measured survivals, Eq. (1) makes opposite predictions, which is the test a criterion has to pass to be worth using (Table 4). On Llama the sum is 1.800 and on DeepSeek 1.018, so both frontier loops should be supercritical; on qwen2.5:7b it is 0.578, so the same loop should decay.

The measured dynamics follow. On Llama the recurrent manager is re-compromised by its own workers’ reports in 0.725 [0.572, 0.839] of runs and does not die out: at round five it is still compromised in 0.700 of runs with the workers near saturation (0.925, 0.975), as $\rho > 1$ predicts; extending the same loop to eight rounds leaves it endemic (manager 0.583, workers 0.900). DeepSeek, whose $\sum_j a_j c_j = 1.018$ sits just above criticality, behaves the same way

Table 4: One criterion, three models, opposite predictions. $\sum_j a_j c_j$ uses the measured per-hop survivals of the same three-agent recurrent pattern; the feed-forward column is the same survivals with the two feedback edges removed.

<i>model</i>	$\sum_j a_j c_j$	ρ <i>rec.</i>	ρ <i>fwd.</i>	<i>prediction</i>
Llama 3.3 70B	1.800	1.342	0.000	supercritical
DeepSeek V3.2	1.018	1.009	0.000	supercritical (marginal)
qwen2.5:7b	0.578	0.760	0.000	subcritical

over eight rounds—the manager settles at 0.567 with workers at 0.767 and 0.700 rather than decaying—so the recurrent verdict is not a single-model artefact. On qwen the same loop decays: the manager’s re-compromise rate falls from 0.775 [0.625, 0.877] to 0.650 and the workers plateau (0.475, 0.525). One criterion, opposite predictions, all borne out.

Two caveats. First, ρ is computed from *isolated* survivals that Sections 5.2–5.7 of the main paper show to be unreliable; the Llama and qwen verdicts survive because neither $\sum_j a_j c_j$ is marginal (1.80, 0.58), whereas DeepSeek’s 1.018 sits close enough to the critical value that its supercritical label should be read together with its measured dynamics (endemic over eight rounds) rather than from the point estimate alone. Second, we study a single recurrent pattern at one temperature; the three models span five to eight rounds, enough to separate decay from saturation, not to demonstrate extinction.